MATH2601 – Higher Linear Algebra

Topic 2 – Vector spaces
Lecture 2.3 – Sums and direct sums

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Related subspaces and subspace sum

Consider the following lemmas about subspaces of any vector space $(V, \mathbb{F})$.

Lemma. If $S \leq V$ and $T \leq V$, then $S \cap T \leq V$.

Proof. Since $\mathbf{0}$ must be an element of both S and T, certainly $S \cap T \neq \varnothing$. Now let $\mathbf{u}, \mathbf{v} \in S \cap T$ and $\lambda \in \mathbb{F}$. Since $\mathbf{u} + \lambda\mathbf{v}$ is a linear combination of vectors in S, it is an element of S, and likewise for T. So $\mathbf{u} + \lambda\mathbf{v} \in S \cap T$, and by the subspace lemma $S \cap T \leq V$.

Lemma. If $S \leq V$ and $T \leq V$, then in general $S \cup T \not\leq V$.

Proof. Consider $S = \left\{ \begin{pmatrix} a \\ 0 \end{pmatrix} : a \in \mathbb{R} \right\}$ and $T = \left\{ \begin{pmatrix} 0 \\ b \end{pmatrix} : b \in \mathbb{R} \right\}$ as subspaces of $\mathbb{R}^2$. Since the union is not closed under addition, $S \cup T \not\leq \mathbb{R}^2$.

Definition. Given a vector space V with $S \leq V$ and $T \leq V$, their subspace sum is given by $S + T = \{\mathbf{s} + \mathbf{t} : \mathbf{s} \in S \text{ and } \mathbf{t} \in T\}$.

Lemma. If $S \leq V$ and $T \leq V$, then $S + T \leq V$.

Proof. Since S and T must be nonempty, certainly $S + T \neq \varnothing$. Now let $\mathbf{u}, \mathbf{v} \in S + T$ and $\lambda \in \mathbb{F}$. Then $\mathbf{u} = \mathbf{s}_1 + \mathbf{t}_1$ and $\mathbf{v} = \mathbf{s}_2 + \mathbf{t}_2$ for some $\mathbf{s}_i \in S$ and $\mathbf{t}_i \in T$. Writing $\mathbf{u} + \lambda\mathbf{v} = (\mathbf{s}_1 + \mathbf{t}_1) + \lambda(\mathbf{s}_2 + \mathbf{t}_2) = (\mathbf{s}_1 + \lambda\mathbf{s}_2) + (\mathbf{t}_1 + \lambda\mathbf{t}_2)$ shows that $\mathbf{u} + \lambda\mathbf{v} \in S + T$, so by the subspace lemma $S + T \leq V$.

Comparing subspaces

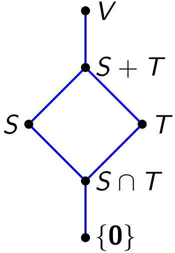
We can conclude for any vector space V with subspaces S and T, that

$$S \cap T \leq S \leq S + T \leq V$$

and

$$S \cap T \leq T \leq S + T \leq V.$$

This is summarised in the Hasse diagram on the right (with the subspace relation $\leq$ as partial order).



Example. Consider the vector space $\mathcal{P}_3(\mathbb{R})$ with subspaces

$$S = \{at^3 + c \,:\, a, c \in \mathbb{R}\} \text{ and } T = \{bt^2 + bt + c \,:\, b, c \in \mathbb{R}\}.$$

Then

$$S \cap T = \{c \,:\, c \in \mathbb{R}\}$$

and

$$S + T = \{at^3 + bt^2 + bt + c \,:\, a, b, c \in \mathbb{R}\}.$$

We can see that each of these is a subspace of $\mathcal{P}_3(\mathbb{R})$, and that $S \cap T \leq S \leq S + T$ and $S \cap T \leq T \leq S + T$.

Note that $\dim(S) = \dim(T) = 2$, $\dim(S \cap T) = 1$, and $\dim(S + T) = 3$.

Subspace sum example

Exercise. Consider the vector space $M_2(\mathbb{R})$ and its subspace S containing all symmetric matrices in $M_2(\mathbb{R})$. Find another subspace $T \leq M_2(\mathbb{R})$ such that $S + T = M_2(\mathbb{R})$. Can you find such a T for which $S \cap T$ is trivial?

Solution. We can write $S = \left\{ \begin{pmatrix} a & b \\ b & c \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$.

A sensible corresponding subspace T might be $T = \left\{ \begin{pmatrix} 0 & d \\ e & 0 \end{pmatrix} : d, e \in \mathbb{R} \right\}$.
Notice in this case that $S + T = M_2(\mathbb{R})$, but $S \cap T = T$ so the intersection is not trivial.

Alternatively, we could choose $T' = \left\{ \begin{pmatrix} 0 & d \\ 0 & 0 \end{pmatrix} : d \in \mathbb{R} \right\}$. Note that in this case $S + T' = M_2(\mathbb{R})$ while $S \cap T' = \{\mathbf{0}\}$, so here the intersection is trivial.

Note that here $\dim(S) = 3$, $\dim(T) = 2$, and $\dim(S \cap T) = 1$, while $\dim(T') = 1$ and $\dim(S \cap T) = 0$. Since $\dim(M_2(\mathbb{R})) = 4$, in particular we can note that $\dim(S + T) = \dim(S) + \dim(T) - \dim(S \cap T) = 3 + 2 - 1$ and that $\dim(S + T') = \dim(S) + \dim(T') - \dim(S \cap T') = 3 + 1 - 0$.

Note we could also have chosen T' to be the set of skew-symmetric matrices (those for which $A^T = -A$) in $M_2(\mathbb{R})$, where the intersection is again trivial.

Dimension of subspace sum

Theorem. If S and T are subspaces of a finite-dimensional vector space $(V, \mathbb{F})$, then $\dim(S+T) = \dim(S) + \dim(T) - \dim(S \cap T)$.

Proof. Let $n = \dim(S \cap T)$ and $B_{S\cap T} = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ be a basis for $S \cap T$. Then since $B_{S\cap T}$ is still linearly independent in S, we can find a basis B_S for S containing $B_{S\cap T}$, where $B_S = \{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{s}_1, \dots, \mathbf{s}_k\}$ and $\dim(S) = n + k$. Similarly, we can find a basis B_T for T where $B_T = \{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{t}_1, \dots, \mathbf{t}_\ell\}$ and $\dim(T) = n + \ell$. So we want to show $\dim(S+T) = n + k + \ell$.

Consider the set $B = B_S \cup B_T = \{\mathbf{u}_1, \dots, \mathbf{u}_n, \mathbf{s}_1, \dots, \mathbf{s}_k, \mathbf{t}_1, \dots, \mathbf{t}_\ell\}$. Using the bases B_S and B_T, any element of $S+T$ can be written as

$$\left(\sum \alpha_i \mathbf{u}_i + \sum \beta_i \mathbf{s}_i\right) + \left(\sum \gamma_i \mathbf{u}_i + \sum \delta_i \mathbf{t}_i\right) = \sum (\alpha_i + \gamma_i)\mathbf{u}_i + \sum \beta_i \mathbf{s}_i + \sum \delta_i \mathbf{t}_i$$

for some $\alpha_i, \beta_i, \gamma_i, \delta_i \in \mathbb{F}$, so B spans $S+T$.

Now suppose $\sum \alpha_i \mathbf{u}_i + \sum \beta_i \mathbf{s}_i + \sum \gamma_i \mathbf{t}_i = \mathbf{0}$ for some $\alpha_i, \beta_i, \gamma_i \in \mathbb{F}$. Set $\mathbf{v} = \sum \alpha_i \mathbf{u}_i + \sum \beta_i \mathbf{s}_i$, so $\mathbf{v} = -\sum \gamma_i \mathbf{t}_i$. Clearly $\mathbf{v}$ is an element of both S (since each $\mathbf{u}_i \in S$) and T, so $\mathbf{v} \in S \cap T$. Then $\mathbf{v} = \sum \delta_i \mathbf{u}_i$ for some $\delta_i \in \mathbb{F}$, so $\sum \delta_i \mathbf{u}_i + \sum \gamma_i \mathbf{t}_i = \mathbf{0}$. Since B_T is linearly independent, $\delta_i = 0$ and $\gamma_i = 0$ for all i. So $\sum \alpha_i \mathbf{u}_i + \sum \beta_i \mathbf{s}_i = \mathbf{0}$, and since B_S is linearly independent, $\alpha_i = 0$ and $\beta_i = 0$ for all i. Thus B is linearly independent.

So B is a basis for $S+T$, meaning $\dim(S+T) = n + k + \ell$ as required.

Dimension of subspaces example

Exercise. Suppose S and T are subspaces of $\mathbb{F}^{13}$ with $\dim(S) = 10$ and $\dim(T) = 12$. Find all possible dimensions for $S \cap T$.

Solution. Since $S \cap T \leq S$, we know that $\dim(S \cap T) \leq 10$.

On the other hand, since $S + T \leq \mathbb{F}^{13}$, we know that $\dim(S + T) \leq 13$. So $13 \geq \dim(S + T) = \dim(S) + \dim(T) - \dim(S \cap T) = 22 - \dim(S \cap T)$, meaning $\dim(S \cap T) \geq 9$.

There are therefore only two cases:

- If $\dim(S \cap T) = 9$, then $\dim(S + T) = 22 - 9 = 13$ which implies $S + T = \mathbb{F}^{13}$.
- If $\dim(S \cap T) = 10$, then $\dim(S \cap T) = \dim(S)$ which implies $S \cap T = S$ and so $S \leq T$.

Direct sums

Definition. If two vector subspaces S and T have trivial intersection (so $S \cap T = \{\mathbf{0}\}$), we can call their subspace sum a direct sum (or internal direct sum) and may denote it as $S \oplus T$.

Note that with this notation, we can write $\dim(S \oplus T) = \dim(S) + \dim(T)$ (but that this only holds if S and T have trivial intersection).

Example. We saw that $M_2(\mathbb{R})$ is the direct sum of the space of symmetric 2×2 matrices, and the space of skew-symmetric 2×2 matrices.

Example. If $\mathbf{u}$ and $\mathbf{v}$ are two non-parallel geometric vectors in $\mathbb{R}^2$, then $\mathbb{R}^2 = \operatorname{span}(\{\mathbf{u}\}) \oplus \operatorname{span}(\{\mathbf{v}\})$.

Example. If $\mathbf{u}_1, \mathbf{u}_2$ and $\mathbf{v}_1, \mathbf{v}_2$ are two pairs of non-parallel vectors in $\mathbb{R}^3$ whose spans are not equal, then $\mathbb{R}^3 = \operatorname{span}(\{\mathbf{u}_1, \mathbf{u}_2\}) + \operatorname{span}(\{\mathbf{v}_1, \mathbf{v}_2\})$, but this is not a direct sum since the two planes intersect in a line.

Example. From the previous example, for at least one of $i \in \{1, 2\}$ we can say that $\mathbb{R}^3 = \operatorname{span}(\{\mathbf{u}_1, \mathbf{u}_2\}) \oplus \operatorname{span}(\{\mathbf{v}_i\})$ since the intersection of the plane and the line is the trivial subspace.

Uniqueness of direct sum decomposition

Lemma. Given subspaces S and T of a vector space V, their sum is direct (that is, $S + T = S \oplus T$) if and only if each element $\mathbf{v} \in S + T$ can be written uniquely as $\mathbf{v} = \mathbf{s} + \mathbf{t}$ for some $\mathbf{s} \in S$ and $\mathbf{t} \in T$.

Proof. For the forward direction, suppose that $S + T = S \oplus T$, that is, that $S \cap T = \{\mathbf{0}\}$. Suppose also that some $\mathbf{v} \in S + T$ can be written in two ways as $\mathbf{v} = \mathbf{s}_1 + \mathbf{t}_1 = \mathbf{s}_2 + \mathbf{t}_2$ for some $\mathbf{s}_i \in S$ and $\mathbf{t}_i \in T$. Then we have $\mathbf{s}_1 - \mathbf{s}_2 = \mathbf{t}_2 - \mathbf{t}_1$. The left-hand side of this equation is an element of S, while the right-hand side is an element of T. So $\mathbf{s}_1 - \mathbf{s}_2 \in S \cap T = \{\mathbf{0}\}$, which implies $\mathbf{s}_1 = \mathbf{s}_2$, and similarly $\mathbf{t}_1 - \mathbf{t}_2 \in S \cap T = \{\mathbf{0}\}$, which implies $\mathbf{t}_1 = \mathbf{t}_2$. So any element $\mathbf{v} \in S + T$ must have a unique sum decomposition.

For the reverse direction, suppose that each $\mathbf{v} \in S + T$ can be written uniquely as $\mathbf{v} = \mathbf{s} + \mathbf{t}$ for some $\mathbf{s} \in S$ and $\mathbf{t} \in T$. Then if $\mathbf{v} \in S \cap T$, in particular $\mathbf{v} \in S$ so $\mathbf{v} = \mathbf{v} + \mathbf{0}$, but also $\mathbf{v} \in T$ so $\mathbf{v} = \mathbf{0} + \mathbf{v}$. Since these sums must be unique, we must have that $\mathbf{v} = \mathbf{0}$. So $S \cap T = \{\mathbf{0}\}$, meaning $S + T = S \oplus T$.

Example using direct sums

Exercise. Show that any $M \in M_2(\mathbb{R})$ can be written uniquely as $M = A + B$ for some $A, B \in M_2(\mathbb{R})$ where $A \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $B \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Solution. Let $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. Let $S = \{A \in M_2(\mathbb{R}) \, : \, A\mathbf{u} = \mathbf{0}\}$ and $T = \{B \in M_2(\mathbb{R}) \, : \, B\mathbf{v} = \mathbf{0}\}$.

Notice that S is nonempty since $\mathbf{0} \in S$, and for any $M, N \in S$ and $\lambda \in \mathbb{R}$ we have $(M + \lambda N)\mathbf{u} = M\mathbf{u} + \lambda(N\mathbf{u}) = \mathbf{0}$. So $S \leq M_2(\mathbb{R})$, and likewise $T \leq M_2(\mathbb{R})$.

Notice that any element $M \in S \cap T$ must satisfy $M \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

Since $\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ is invertible, the only solution is $M = \mathbf{0}$. So $S \cap T$ is trivial, meaning the sum of S and T is direct.

Furthermore $\dim(S \oplus T) = \dim(S) + \dim(T) = 4 = \dim(M_2(\mathbb{R}))$, so we must have that $S \oplus T = M_2(\mathbb{R})$. Thus the uniqueness of sum decomposition is guaranteed by the direct sum property.

Complementary subspace

Lemma. If S is a subspace of a vector space V, then there exists some $T \leq V$ for which $V = S \oplus T$.

Proof. Let $\{\mathbf{s}_1, \dots, \mathbf{s}_k\}$ be a basis for S. Then we can minimally extend it to a basis of V given by $\{\mathbf{s}_1, \dots, \mathbf{s}_k, \mathbf{t}_1, \dots, \mathbf{t}_\ell\}$ for some $\mathbf{t}_i \in V - S$. Set $T = \operatorname{span}(\{\mathbf{t}_1, \dots, \mathbf{t}_\ell\})$. Certainly $S + T = V$ since any element of $S + T$ is a linear combination of $S \cup T$, which is a basis for V. Furthermore, by construction $S \cap T = \{\mathbf{0}\}$. So T is a subspace of V for which $S \oplus T = V$.

Definition. If S is a subspace of a vector space V, then any (not usually unique) $T \leq V$ such that $V = S \oplus T$ is called a complementary subspace for S in V.

Example. For any nonparallel geometric vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$, a complementary subspace for $\Pi = \operatorname{span}(\{\mathbf{u}, \mathbf{v}\})$ is the line $\operatorname{span}(\{\mathbf{w}\})$ for any $\mathbf{w} \in \mathbb{R}^3$ not parallel to the plane Π.

External direct sum

We previously referred to the direct sum $S \oplus T$ as an internal direct sum. This is really an adaptation of the more general external direct sum.

Definition. The external direct sum of two vector spaces $(U, \mathbb{F})$ and $(V, \mathbb{F})$ is the vector space formed by taking their Cartesian product, namely $U \times V = \{(\mathbf{u}, \mathbf{v}) : \mathbf{u} \in U, \mathbf{v} \in V\}$, and equipping it with the operations of vector addition given by $(\mathbf{u}_1, \mathbf{v}_1) + (\mathbf{u}_2, \mathbf{v}_2) = (\mathbf{u}_1 + \mathbf{u}_2, \mathbf{v}_1 + \mathbf{v}_2)$ and scalar multiplication given by $\lambda(\mathbf{u}_1, \mathbf{v}_1) = (\lambda\mathbf{u}_1, \lambda\mathbf{v}_1)$ for all $(\mathbf{u}_1, \mathbf{v}_1), (\mathbf{u}_2, \mathbf{v}_2) \in U \times V$ and $\lambda \in \mathbb{F}$. This vector space is denoted $U \oplus V$.

Notice that with this definition, the external direct sum between two vector spaces U and V is the sum $U \oplus V = (U \oplus \{\mathbf{0}\}) + (\{\mathbf{0}\} \oplus V)$, where clearly $(U \oplus \{\mathbf{0}\}) \cap (\{\mathbf{0}\} \oplus V) = \{(\mathbf{0}, \mathbf{0})\}$ is trivial, hence justifying our internal direct sum notation.

In actuality, for subspaces S and T of V, the spaces $S + T$ and $S \oplus T$ (as an external sum) contain very different types of elements, but have exactly the same structure (since we know each element of $S + T$ has a unique sum decomposition $\mathbf{s} + \mathbf{t}$ for $\mathbf{s} \in S$ and $\mathbf{t} \in T$). So when we write $S + T = S \oplus T$, what we really mean is $S + T \cong S \oplus T$ for some notion of isomorphism over vector spaces. We will better define this in the next section!